SYSC 4906 G

Methodologies for Discrete Event Modeling and Simulation

Pollinator Foraging Simulation

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GitHub: https://github.com/GAchuzia/Cell-DEVS-Bee-Foraging

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Introduction

In order to better understand plant-pollinators interactions and how they impact their environment, we are extending the work submitted in Assignment 2 by adding onto the bee nectar foraging system. For our term project, we will include multiple pollinators, plant species interactions, and more environmental factors.

Asymmetry is introduced by modifying the neighborhood topology of selected cells. In our water-barrier scenario, adjacency links are removed between specific cell pairs, creating an impassable river segment without changing the local transition rule.. To make this possible, we first modified our previous system to incorporate the bees' agentic DEVS model, and then expanded to adapting the terrain and adding a plant species once the agentic bees were fully functional. Finally, once a solid system was in place, we introduced competition against the bees with a new butterfly model, extra plant species, and added complexity to the environment.

Background

This project was originally developed and implemented in CD++ for our group's Assignment 2 of this course. It is based on a paper titled *Plant–pollinator interaction model with separate pollen and nectar dynamics*, which studies the mutualism between plants and pollinators. The paper explores that most pollinators are generalist foragers meaning that they can extract pollen from any plant, whereas the plants themselves can only use their own species' pollen to reproduce. This is described as inter-specific pollen (IPT) transfer which can compromise species with a weaker presence in the environment and deplete their population over time. By generalizing the paper's mathematical formulas on nectar and pollen dynamics as well as the propagation rates modeled by constants, we managed to recreate a simple model that showcases pollen/nectar transfer via bee movement from one nectar patch to the next. For this extension of our second assignment, we will focus more upon the interference caused by IPT on plant fitness with our newly introduced species of plants and butterflies as well as added environmental factors, with the goal of demonstrating how the plant to pollinator dynamics can move from a mutualistic relationship to a parasitic one with respect to the species' traits and the ecological context of the environment.

Overview of the Paper

The key contribution of this paper is the separation of nectar and pollen dynamics into two distinct but interconnected processes. Nectar represents the resource consumed by pollinators, while pollen represents the reproductive mechanism of plants. Unlike nectar, which is consumed and removed from the system, pollen undergoes a two-step process involving pickup by pollinators and subsequent delivery to plant stigmas.

A central innovation of the model is its treatment of pollen transfer efficiency, particularly the role of inter-specific pollen transfer (IPT). Because many pollinators visit multiple plant species, pollen may be delivered to the wrong species, reducing reproductive success. This introduces the possibility that plant–pollinator interactions are not always purely beneficial. Instead, they can vary along a spectrum from mutualism (both species benefit) to antagonism (pollinators benefit while plants are harmed) depending on ecological conditions.

The model also incorporates resource limitations and saturation effects, meaning that plant benefits do not increase indefinitely with pollinator abundance. This produces more realistic dynamics compared to earlier models and prevents unbounded population growth. Additionally, the model predicts Allee effects, where plants at low population densities experience reduced reproductive success due to insufficient pollen transfer between conspecific individuals.

Another important feature of the model is its emphasis on ecological context. The outcome of plant–pollinator interactions depends on the presence of alternative plant and pollinator species, as well as differences in their efficiency as nectar sources or pollen carriers. As a result, the same interaction can shift between cooperative and competitive outcomes depending on the surrounding community.

Finally, while the model focuses on pollen as a reproductive resource, the author acknowledges that pollen is also consumed by pollinators as a food source, and suggests this as a potential extension to the model rather than including it in the core formulation.

Overall, the paper provides a more detailed and flexible framework for understanding plant–pollinator systems by explicitly modeling the mechanisms of resource exchange and pollen transfer, highlighting how local interactions can lead to complex and context-dependent ecological dynamics.

Original Model from Assignment 2

In Assignment 2, we implemented a Cell-DEVS model to simulate a bee nectar foraging ecosystem. Each cell stored the local levels of nectar and pollen, as well as the number of bees currently occupying that patch. The simulation followed rules for how bees move between cells in response to resource availability, and how nectar and pollen grow, decay, and are consumed over time, all within a symmetric 10×10 grid using a Von Neumann neighbourhood.

The model tracked three state variables per cell: nectar level (0–100), pollen level (0–50), and bee count (0–50). Cells behaved according to the following rules:

- Bees depart a cell when local resources fall too low, or when a neighbouring cell offers significantly more resources

- Bees are attracted into a cell from neighbours when that cell has higher nectar levels, with movement scaled by the size of the difference
- Nectar regrows each step, boosted slightly by local pollen, and is consumed by bees at a saturating rate that prevents total depletion
- Pollen is transported between cells by travelling bees, and also undergoes mild autonomous regrowth and decay

To run the simulation, we created custom JSON configuration files that defined the initial state of every cell, including nectar and pollen levels, bee counts, grid topology (wrapped or unwrapped), and colour gradient mappings for visualization. Four test scenarios were produced to validate the model: a no-bee baseline, a center burst of activity, and corner-placed resource clusters on both wrapped and unwrapped grids. The simulation output was visualized using the Cell-DEVS Web Viewer, allowing us to observe how bee populations and resource levels co-evolved across the grid over 50 time steps.

Design Modifications for the Term Project

Building on the Assignment 2 foundation, the term project introduces several layers of complexity to better capture the dynamics of real plant-pollinator ecosystems. The modifications were implemented incrementally in three phases, each adding new components only once the previous phase was stable and validated.

Hybrid Agent-DEVS

Hybrid Agent-DEVS means combining the atomic DEVS models for mobile pollinators (bees and butterflies) and a Cell-DEVS grid for patch dynamics. The term project couples atomic pollinator agents to a Cell-DEVS nectar grid in Cadmium. Each Bee and Butterfly model extends Atomic<State> with its own time advance, internal transition, external transition, and output function. The pollinators send structured messages (e.g. BeeMovement/ButterflyMovement) that include an identifier, a consumption request, and pollen type label to classify different and same-species on a patch.

Bee Roles

Bees can have varying colony roles; in foraging, a key distinction is between scouts and recruits/foragers [2]. In our model, we represent two role classes (forager and scout) through different extraction and capacity parameters. A forager is configured with the highest nectar extraction rate, and the largest storage limit, making it the primary resource gathering role. The scout has a lower extraction rate and a small capacity, reflecting its limited direct harvesting and lighter exploratory function [2]. Functionally, the transition logic best highlights the foragers unique behavior; only foragers accumulate carried nectar from cells. This means capacity-driven movement is mainly triggered by foragers, while scouts and nurses have less influence over grid changes.

Butterfly Roles

Butterflies are modeled as a separate atomic pollinator class, with behavior distinct from bees. These pollinators have a less selective foraging strategy and are more inaccurate pollinators. In the cell rules, when butterflies visit flowers, they are more likely to leave the wrong pollen type (heterospecific pollen) and less likely to leave the matching pollen type (conspecific pollen) for a given plant species. This matters because plants reproduce best when they receive matching pollen [3]. So even though butterflies help move pollen around, they can also reduce plant reproductive success by increasing conflicting pollen types. At the same time, butterflies consume nectar so they also compete with bees for the same food source [3].

Asymmetric Cell-DEVS Integration

Terrain heterogeneity is modeled using an asymmetric Cell-DEVS topology. The default landscape uses a von Neumann neighborhood, but cells can override that default with a cell-specific neighborhood list. In Test 6, a full-row river barrier is created: all ten cells in row 4 (0,4) through (9,4) are assigned an empty neighborhood, meaning they observe no neighbors at all. Symmetrically, cells in rows 3 and 5 do not include any row-4 cell in their own neighborhood lists. This completely severs interaction across row 4 in both directions, splitting the grid into two independent halves, while all local transition rules remain unchanged.

Multi-Species Competition

Competition is represented though shared resource pressure between pollinators. Bees and butterflies consume from the same nectar stock, and deposits are evaluated against two nondescript plant species to separate same species and different species pollen accumulation. In mixed plant species landscapes, overlap in pollinators visiting paths increases cross-species pollen exposure, which can reduce effective nectar reproduction (and plant reproduction) [3].

Bee to Butterfly Competition

Bee-butterfly competition is indirect but persistent since both pollinators target the same patches, so occupancy overlap increases nectar depletion. Because bees and butterflies have varying speeds and consumption rates, they do not contribute equally to outcomes. In mixed pollinator runs (seeTest 5), butterflies tend to increase incorrect pollen transfer relative to bee-dominant regions.

Model Structure and Coupling Scheme

This section defines the structure of the atomic and coupled models in the Pollinator Foraging System, using a hybrid composition in Cadmium where mobile pollinator agents (atomic DEVS) interact with a nectar landscape (Cell-DEVS grid). The top-level model couples one NectarGrid component with pollinator atomics (Bee, Butterfly) and routes agent outputs into grid input ports.

Atomic Cell-Devs

$$\text{IDC} = \langle X, Y, S, \theta, I, N, d, \tau, \delta_{int}, \delta_{ext}, . \lambda, D\rangle$$

X (Input Set)

For each cell the input contains two external pollinator event types plus neighborhood state messages.

$$x_b = \langle \text{bee\_id, x, y, entering, consumption\_request, pollen\_type} \rangle$$

Bee movement event

- bee_id (int): bee identifier
- x,y (int): target coordinate
- action (int): movement status code (0 = leaving, 1 = arriving, 2 = staying)
- consumption_request (double): requested nectar extraction
- pollen_type (int): pollen identity carried by bee
- role (Role): bee behavior type (forager and scout)

Butterfly movement event

$$x_f = \langle \text{butterfly\_id, x, y, entering, consumption\_request, pollen\_type} \rangle$$

- Same semantics as bee, excluding roles

Y (Output Set)

Each cell outputs its current state to neighbors/logs.

$$y = \langle \text{nectar, pollen, bees, butterflies, plant\_species, pollen\_type, conspecific\_pollen, heterospecific\_pollen} \rangle$$

- nectar (double): nectar level
- pollen (double): pollen level
- bees (int): local bee occupancy
- butterflies int): local butterfly occupancy
- plant_species (int): resident plant species
- conspecific_pollen (double): same-species (conspecific) pollen count
- heterospecific_pollen (double): different species/incompatible pollen count

S (State Set)

$$S = \langle N, P, B, F, PS, PT, CP, HP \rangle$$

- N: nectar level
- P: pollen level
- B: number of bees
- PS: plant species
- PT: current pollen identity
- CP: conspecific pollen cumulative value
- HP: heterospecific pollen cumulative value

θ (Output Delay)
The cell uses a fixed output delay of 1 simulation time unit (outputDelay = 1.0).

$$\theta = 1.0$$

I (Cell Identifier)

$$I = (i,j), \quad i,j \in \{0, \ldots, 9\}$$

N (Neighborhood)
- Asymmetric Neighborhoods
- Von Neumann Neighborhoods

d (Cell Delay Type)
- 1 time unit

***τ* (Local Computing Function)**
The local transition function τ determines the next state of each cell based on its current state and the states of its neighboring cells.

For each cell (0,0), the updated values are computed as follows:
- $N'_{0,0} = N_{0,0} + R(P_{0,0}) - C(B_{0,0})$
- $P'_{0,0} = P_{0,0} + \sum$ incoming_pollen $- \sum$ outgoing_pollen
- $B'_{0,0} = B_{0,0} + \sum$ incoming_bees $- \sum$ outgoing_bees

Where:
- $R(P_{0,0})$ represents nectar regeneration influenced by local pollen levels
- $C(B_{0,0})$ represents nectar consumption based on bee activity
- incoming_pollen and outgoing_pollen depend on pollinator movement between neighboring cells
- incoming_bees and outgoing_bees are determined by relative nectar levels between the current cell and its neighbors

The function enforces the following constraints:
- $0 \leq N'_{0,0} \leq Nmax$
- $0 \leq P'_{0,0} \leq Pmax$

- $B'_{0,0} \geq 0$

δ_{int}, (Internal Transition)
Internal evolution applies τ, then restricts values to valid ranges.

δ_{ext}, (External Transition)
Upon receiving bee/butterfly events:

1. Update occupancy using entering
2. Subtract nectar by consumption_request
3. Update pollen-related values
4. Classify deposited pollen:
   a. if pollen_type = plant_species -> CP increases
   b. else -> HP increases
5. Read neighborhood messages and refresh neighbor states used by local logic

λ (Output Function)
Outputs current state vector to neighborhood/log stream.

D (Default State)
The default cell state (from nectarState defaults/JSON defaults).

$$\langle 10.0, \mathit{5.0, 0, 0, 1, 0, 0.0, 0.0} \rangle$$

Coupled Cell-Devs

$$CM = \langle I, X, Y, X_{list}, \mathrm{Y}_{\mathrm{list}}, \eta, N\{m, n\}, C, B, Z, select \rangle$$

I (Input Interface)
Grid-level external inputs:

- In_bee_move (type BeeMovement)
- In_butterfly_move (type ButterflyMovement)

X (Set of Input Values)

- X: same event/state types as atomic Cell-DEVS input set

Y (Set of Output Values)

- Y: cell neighborhood output states used by Cell-DEVS logging

X_{list} (List of External Inputs)

- {in_bee_move, in_butterfly_move} at grid level

Y_{list} (List of External Outputs)

- Standard cell neighborhood outputs

η (Neighborhood Size)

- Asymmetric Neighborhoods (e.g. rivers)
- Von Neumann Neighborhood

N (Relative Neighbor Position)

- N = {(-1, 0), (1, 0), (0, -1), (0, 1)}

{m, n} (Grid Size {row, column})

- {10, 10}

C (Set of All Cells)

- $C=\{C_{ij} \mid i \in [0,9],\ j \in [0,9]\}$

B (Border Condition)

- Wrapped = true (tordial)
- Wrapped = false (non-wrapped)

Z (Coupling Relation)

Top-level couple model (BeeSimulation)

- Bee_1.out -> NectarGrid.in_bee_move
- Bee_2.out -> NectarGrid.in_bee_move
- $Butterfly_1$.out -> NectarGrid.in_butterfly_move

Within NectarGrid

- in_bee_move -> every cell in_bee_event
- in_butterfly_move -> every cell in_butterfly_event
- Neighborhood exchange among cells according to N

select (Tie-Breaking Function)

Default simulator scheduling/tie handling.

Asymmetric Specification

$$C_i = \langle X_i^N,\ X_i^E,\ Y_i,\ S_i,\ N_i,\ \tau_i,\ d_i,\ delay_i \rangle$$

S_i:

$$S = \langle N, P, B, F, PS, PT, CP, HP \rangle$$

- N: nectar level
- P: pollen level
- B: number of bees

- PS: plant species
- PT: current pollen identity
- CP: conspecific pollen cumulative value
- HP: heterospecific pollen cumulative value

N_i: Cell Neighbourhood is cell-specific and may differ because of the river barrier

Neighborhood

- Ni is cell-specific and may differ due to environmental constraints.
- By default, cells use a Von Neumann neighborhood:
- Ni = {(i+1,j), (i−1,j), (i,j+1), (i,j−1), (i,j)}

For river cells, adjacency is modified such that:

- all cells (x,4) for x $\in$ {0,…,9} are assigned an empty neighborhood (they observe no neighbors),
- cells in rows 3 and 5 do not include any row-4 cell in their neighborhood lists, making the barrier fully bidirectional and spanning the entire grid width.

Input Sets:

- Xi^N: neighborhood state inputs from adjacent cells
- Xi^E: external pollinator events (bee and butterfly movement)

Output Set:

- Yi: cell state vector broadcast to neighbors and logging system

Local Transition Function:

- τi applies nectar regeneration, pollen transfer, and pollinator movement rules as defined in Section X.

Delay Function:

- di(s) = 1

Delay Type

- delay = inertial

Demonstrating Simulation

This section presents the configuration files, visualization setup, and simulation outputs for the Pollinator Foraging system. Each test scenario is designed to isolate or combine one or more of the new environmental and biological features introduced in the term project, complemented by the four baseline tests from Assignment 2.

Visual Guide to Simulation Output

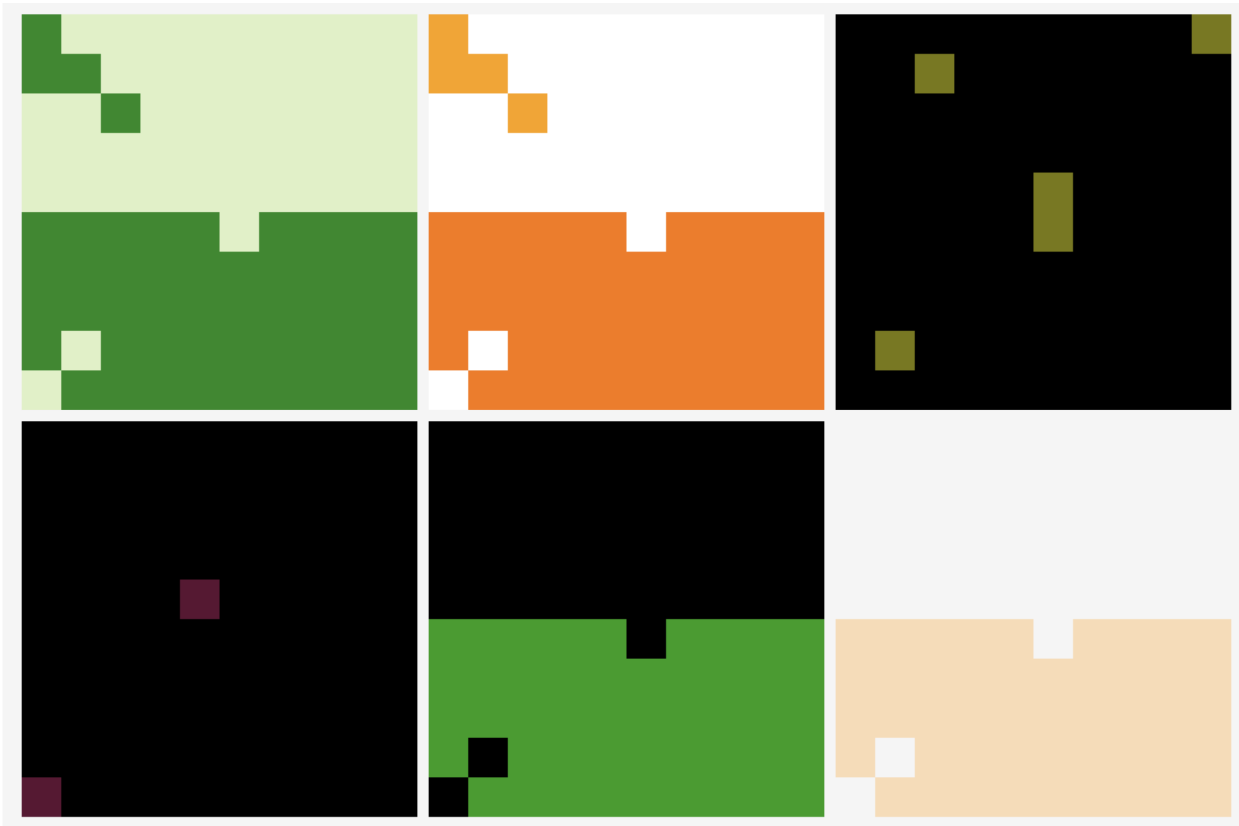

Figure 1: Cell-DEVS View of Pollinator Foraging Simulation

The grid is visualized in 6 panels (left to right, then top to bottom); nectar level, pollen level, number of bees, number of butterflies, plant species, and mismatched pollen level.

Nectar Panel:
- ⬜White to 🟩Green gradient
- Bins: 0-10, 10-25, 25-40, 40-60, 60-80, 80-100

Pollen Panel:
- ⬜White to 🟨Yellow to🟧Orange gradient
- Bins: 0-5, 5-10, 10-20, 20-30, 30-40, 40-50

Bees Panel:
- ⬛Black to 🟨Yellow to 🟥Red gradient
- Bins: 0-0.1, 0.1-1, 1-3, 3-8, 8-15, 15-25

Butterflies Panel:
- ⬛Black to 🟥Dark Pink to Light Pink

Plant Species Panel:
- No plants is ⬛Black Plant A is 🟩Green Plant B is Blue 🟦

Mismatched (heterospecific) Pollen Panel:

- ⬜ White to 🟧Orange to 🟫Brown
- Bins: 0-1, 1-3, 3-6, 6-10, 10-20

Model Behavior

Resources

Each patch tracks nectar (0-100) and pollen (0-50). Nectar grows faster when pollen is already present, decays with its own level, and drops extra when bees or butterflies sit on the cell (butterflies consume more nectar than bees). Pollen decays, regrows slowly below its cap, and the pollen type clears when pollen is nearly gone.

Pollinators

Bees and butterflies are modeled as separate atomic agents that send movement and feeding events into the cell grid. When a pollinator arrives or leaves, the cell adjusts the corresponding count (bee/butterfly count for that cell). When a consumption request is positive, nectar drops and pollen receives a fraction of the withdrawn nectar (bees and butterflies use different conversion factors, so the same nectar draw does not imply the same pollen gain). When a visitor carries a non-zero pollen type, the cell updates conspecific vs heterospecific accumulation depending on whether the pollen type matches the plant species (Plant A has pollen type A, Plant B has pollen type B).

Environmental Factors

Environments in which pollinators live vary, which affects their access to pollen and their population growth. We took account of barriers that are in the environment such as a river. The chances of pollinators travelling across this river are low as pollinators rely on local cues like scent to locate plants. Environmental conditions are not uniform and are therefore represented by asymmetric cells.

Plant Species

Two plant species, A and B were considered in the environment. Each species produces its own type of pollen (A and B) which influences the behavior of pollinators. The availability of each pollen type affects pollinator foraging patents which impacts growth and distribution of pollen and nectar within the system.

Experimental Frame

This section extends the testing framework developed in Assignment 2 to evaluate the term project model. We revisit the original baseline scenarios using the updated implementation to verify continuity and highlight any behavioral changes introduced by the new dynamics. We then introduce multi-pollinator/species interaction and stronger competition conditions.

Regression Tests from Assignment 2 (Updated Model)

Test 1: Baseline Resource Regeneration (No Pollinators)

This test sets all cells to a homogeneous baseline with no active pollinators in motion. At t = 0, the cells are {nectar: 10, pollen: 5, bees: 0, butterflies: 0}. Under these conditions, resource evolution is governed entirely by the resources' regeneration and decay.

- Simulation time does not progress in this test because there is no update the localComputation function

Test 2: High Activity Patch

The center patch in this test starts with much higher activity and higher resources than the rest of the grid. Since nectar loss is proportional to local pollinator count, the center cells experience a much faster depletion in nectar compared to outside the center of high activity. As simulation time progresses, center cells experience faster nectar depletion than the outer cells due to their high occupancy and consumption by pollinators.

Results show that:

- High bee and butterfly density causes stronger resource depletion
- The new model responds correctly to the initial conditions from Assignment 2

Test 3: Corner Burst Without Wrapping

This test moves the high-activity patch to the corner and runs with non-wrapped borders. At t = 0, corner cells change slowly but since there are only two neighbours, it is only able to expand in that direction. The high population is restricted by the boundary which acts like a barrier.

Results shows that:

- Bees can be restricted by barriers but also learn how to adapt to them
- Wrap around enabled, it leads to fast and more widespread activation

Test 4: Corner Burst With Wrapping

This test shows a grid with a wrap-around boundary and how bees interact with cells on the opposite side instead of hitting the boundary. This allowed activity to spread across all edges simultaneously creating a symmetric diamond shape.

Results shows that:

- Bees at boundary cells interact opposite edge creating activity all 4 corners which validates the coordinate wrapping
- There were high nectar levels in the corners while center was thinning which shows migrations occurs to area where supply meets demand

New Tests for Multi-Pollinator & Multi-Species Dynamics

Test 5:
This test introduces both bees and butterflies into the same environment with overlapping plant species. The goal is to observe how competition for nectar resources and differences in pollination efficiency affect overall system dynamics.

In this scenario, bees and butterflies are initialized in separate regions and gradually interact as they move across the grid. Bees, which are more efficient pollinators, tend to maintain higher levels of conspecific pollen (CP), while butterflies contribute more to heterospecific pollen (HP) due to their less selective behavior.

Results show that:

- Nectar depletion occurs faster in regions with both pollinators
- Butterfly presence increases mismatched pollen accumulation
- Plant reproduction efficiency decreases in areas dominated by butterflies

This demonstrates how introducing pollinators can shift the system away from mutualism and toward competition or even parasitism.

Test 6:
This test evaluates the impact of environmental constraints on pollinator movement and resource distribution. A full-row river barrier is introduced using asymmetric Cell-DEVS topology: all ten cells in row 4 are assigned empty neighborhoods and are excluded from the neighborhood lists of rows 3 and 5, effectively splitting the entire grid into two fully independent halves.

All ten row-4 cells have no neighbors; rows 3 and 5 cells do not observe row 4, creating a complete and impassable barrier across the full grid width without modifying any local transition rule.

Key Findings and Limitations

Our key finding for this project was the successful hybrid development of a Cell-DEVS and DEVS model working together to simulate bee foraging. By having the bee and butterfly as atomic models operating on a grid of nectar cells, we are able to see how pollinators interact dynamically with the environment. Our simulation included two different types of pollinators and two different plant species competing for nectar, as well as an asymmetric Cell-DEVS topology implementing a full-row river barrier (Test 6), in which all cells in row 4 are topologically isolated, demonstrating that spatial connectivity alone can partition emergent pollinator and resource dynamics into fully independent halves of the grid without modifying any local transition rule.

The remaining limitations were in features we did not have time to incorporate. We did not use live environmental data to simulate the environment; while our parameters were grounded in research, real-time data would improve fidelity. We also explored but did not finish a static beehive model where bees could deposit pollen at the hive.

Conclusion

This report presented an extended Cell-DEVS simulation of pollinator foraging dynamics, built upon the single-species bee nectar model developed in Assignment 2. The core contribution of the term project is the integration of three interacting extensions: multi-pollinator dynamics, multi-species plant competition, and asymmetric environmental structure.

The inclusion of butterflies alongside bees demonstrates how differences in pollination efficiency can significantly alter system behavior. While bees promote effective pollen transfer and support plant reproduction, butterflies increase heterospecific pollen deposition, reducing plant fitness through increased inter-specific pollen transfer and introducing competitive effects. This highlights that plant–pollinator interactions are not purely mutualistic, but instead depend on the ecological context.

Additionally, the use of asymmetric Cell-DEVS to model environmental constraints, such as the river barrier, shows how modifying spatial connectivity can influence emergent patterns. Regions become partially isolated, leading to uneven resource distribution and different local dynamics across the grid.

Overall, the simulation demonstrates how simple local rules and interactions can generate complex system-wide behavior. Despite its simplified assumptions, the model effectively captures key ecological dynamics and demonstrates the effectiveness of Cell-DEVS for modeling spatially distributed ecological systems with interacting agents and heterogeneous environments.

References

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